GOWTHAM U

B.E. - Mechatronics
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Profile:

A results-driven Mechatronics Engineer passionate about robotics, computer vision, and autonomous systems. Experienced in developing intelligent automation solutions by integrating cutting-edge technologies such as AI, robotics, and ROS2. Successfully completed an internship at TCS Innovation Lab, where I worked on real-world robotic applications combining edge computing, AI, and ROS-based systems. Eager to contribute to forward-thinking teams focused on building impactful automation and smart systems.

Education:

Rajalakshmi Engineering College *B.E Mechatronics Engineering*	*CGPA: 7.83*	2022 – 2026 *Chennai, Tamil Nadu*
EDUSTAR INTERNATIONAL SCHOOL *HSC*	*83.2%*	2021 – 2022 *Kovillpatti, Tamil Nadu*
EDUSTAR INTERNATIONAL SCHOOL *SSLC*	*83%*	2019 – 2020 *Kovilpatti, Tamil Nadu*

Projects:

Autonomous EV Car Conversion (Ongoing)

- Collaborating with seniors to upgrade a custom-built electric vehicle (EV) into an autonomous car.
- Integrating ROS2 with AI-based perception and planning systems for autonomous driving capabilities.
- Utilizing sensor fusion with LiDAR, cameras, and GPS-IMU for environment awareness and localization.
- Implementing modules for autonomous navigation, object detection, and lane following.
- Focused on creating a full-stack robotic architecture involving edge processing and real-time decision-making.

Autonomous Bot ***(*** GitHub***)***

- Designed and developed an autonomous mobile robot with LiDAR-based perception.
- Implemented Simultaneous Localization and Mapping (SLAM) and Nav2 for autonomous navigation.
- Integrated real-time obstacle detection and path planning algorithms using ROS 2.

Number Plate Recognition System *(Ongoing)*

- Developed an automatic number plate recognition (ANPR) system using YOLO and OpenCV.
- Implemented image preprocessing techniques, including grayscale conversion, thresholding, and noise reduction.
- Used deep learning-based OCR for accurate character segmentation and recognition.
- Designed and tested the system for real-time vehicle identification in traffic environments.

Humanoid Robot and Control App

- Built a humanoid robot using an ESP32 microcontroller and servo motors for precise movement.
- Developed an Android control app for remote operation of the humanoid robot.
- Implemented gesture-based control and human-like movements for enhanced interaction.

AI/ML-Based Automated PID Tuning

Control Systems | Robotics | AI/ML

- Developed an automated PID tuning system using machine learning to reduce manual trial-and-error tuning
- Trained a model to optimize Kp, Ki, Kd based on system response parameters (error, overshoot, settling time)
- Improved control performance and stability for robotics/drone control applications
- Integrated classical PID control with data-driven optimization techniques

Technical Skill:

Programming Languages:

- Python
- C & C++

Robotics & Simulation:

- Robot Operating System (ROS/ROS2)
- Gazebo Simulation
- SLAM & Navigation (Nav2)

Computer Vision & AI:

- OpenCV for real-time image processing
- YOLO for object detection

Software & Development Tools:

- Visual Studio
- SolidWorks
- MATLAB

Competitions:

Electric Bike Design Challenge (EBDC'25) – Season 5.0
Innovation Team Lead | Sept 2025 | Sri Ramakrishna Institute of Technology, Coimbatore

- Led the development and implementation of a new innovation in the electric bike
- Gained hands-on experience in EV systems, mechanical fabrication, and endurance testing
- Team secured 5 awards: Best Business Plan & Cost, Best Aesthetic & Ergonomics, Best Team Spirit, Best Fair Play, Runner-up – Endurance Test

E-Yantra Robotics Competition (eYRC) 2025–26

Stage 2 Qualifier | Ongoing

- Qualified up to Stage 2 in a national-level robotics competition organized by IIT Bombay
- Selected for and received a free official drone development kit for advanced robotics tasks
- Currently working on drone-based robotics challenges, focusing on control, integration, and system-level problem solving
- Gaining exposure to real-world robotics workflows, documentation, and evaluation standards

Achievements:

Technical & Placement Mentor (Volunteer)
Rajalakshmi Engineering College | Dec 2025

- Volunteered as a technical and placement mentor for juniors during the Winter Camp, guiding them in fundamentals, project direction, and placement preparation
- Supported juniors with technical concepts, interview preparation, resume guidance, and peer mentoring in collaboration with teammates

Internship:

TCS Innovation Lab – Robotics PoC Intern
Chennai, Tamil Nadu | [2-Month]

- Contributing to cutting-edge Proof-of-Concept (PoC) projects in the field of robotics and industrial automation.
- Developed and tested ROS/ROS2-based robotic systems integrated with AI for automation in real-world environments.
- Worked with sensors and frameworks such as Gazebo and Nav2 for simulation and navigation.
- Gained valuable exposure to corporate work culture, team collaboration practices, and professional software development workflows.
- This internship significantly enhanced my technical and communication skills, and gave me a clear insight into the expectations and dynamics of the tech industry.